ANIMESH RAJ | 23IE10004

INSTRUMENTATION ENGINEERING (B.Tech 4Y)

+91 7004140025 animeshraj958@gmail.com wildcraft958 animeshraj958

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	**8.17 / 10**
2023	AISSCE (CBSE Class-XII)	Delhi Model Public School, Patna	94.2%
2021	AISSE (CBSE Class- X)	Bishop Scott Boys' School, Patna	98%

INTERNSHIPS

Research Intern | AI-Driven Medical Form Automation Pipeline Using LLMs | Prof. Debdoot Sheet **Mar 2025 - Present**

- An end-to-end AI system was designed to semi-automate medical prescription forms by translating HTML structures into JSON format and using **LLMs** for smart form filling, thus solving fundamental bottlenecks in healthcare workflows in resource-poor clinical settings
- Utilized strong pipeline with **LangChain, Ollama Llama3**, and **BeautifulSoup** libraries for efficient form processing and conversational AI integration, creating scalable solution deployed in remote **SSH environment** with voice interface ability for vernacular input processing
- The module-based design includes enhanced management of chat history and context-aware response generation. It includes a validation review interface and exemplifies real-world application of **LLMs** in filling the gaps in healthcare accessibility for rural disadvantaged individuals

PROJECTS

Vehicle and Pedestrian Counter | Self Project Link **May 2025**

- Developed modular Python pipeline using **YOLOv8** object detection and **SORT** tracking algorithm to automatically detect and count multiple vehicle classes (cars, buses, trucks, motorcycles) and pedestrians in urban traffic footage with mask-based ROI filtering for enhanced accuracy
- Implemented real-time visualization system with **OpenCV** overlays displaying bounding boxes, unique track IDs, and live class tallies, integrated line-crossing logic using centroid tracking to ensure precise per-object counting and prevent duplicate counts during occlusion scenarios
- Achieved over >**20 FPS** inference speed on various video inputs and validated detection accuracy in different lighting conditions using pretrained **COCO** weights. The **CSV** export functionality allows for seamless traffic analysis and reporting which can be used for further analytics of traffic

EdTech Growth Strategy Dashboard | Data Analyst Project **Apr 2025**

- Analyzed a dataset of over **10,000** online courses from Kaggle to identify trends by category using Python and **Power BI**. Standardized course durations to allow for flexible scheduling and created an **actionable dashboard** to guide content expansion in high-demand categories.
- Developed dynamic visualizations that effectively compared **course formats, viewer engagement, and language preferences**. Leveraged **DAX** for advanced dynamic filtering. Successfully identified the **top 5 languages** in each category and pinpointed high-engagement courses
- Evaluated viewer engagement drivers by ranking top instructors, analyzing duration and skill correlations, and measuring subtitle impact. Found that subtitles increased views for technical courses by **15%** and identified optimal course lengths for various subjects.

Vision Transformer (ViT) Implementation from the ICLR'21 Paper | Self Project Link **Mar 2025**

- Applied the whole Vision Transformer architecture of "An Image is Worth 16x16 Words" paper with **PyTorch**, including image patch tokenization, positional encoding, and multi-head self-attention mechanisms for visual data processing on image classification tasks
- Built end-to-end training pipeline on **CIFAR-10** dataset using patch embedding layers, transformer encoder blocks, and classification heads, showcasing practical application of attention mechanisms outside the usual NLP applications to computer vision tasks
- Understanding transformer generalization across modalities involved patch-based image processing and self-attention or multi-head attention computations. This provided insights into how transformers manage visual data representation and feature learning.

COMPETITION/CONFERENCE

Data Analytics General Championship 2025 | Bronze Winning Team | IIT Kharagpur Link **Mar 2025**

- Designed a **Marketing Mix Modeling pipeline** using Meta's Robyn framework, which integrated geometric adstock and Hill saturation curves across three channels. This approach resulted in a **31.9% revenue uplift** through the optimized allocation of an **$18.7 million annual budget**
- Developed a **response decomposition waterfall analysis** and predictive models, validated by comparing actual revenue to predicted revenue, along with correlation hypothesis testing. This ensured model accuracy with a significance level of **p - value ($p < 0.05$)**
- Conducted **customer behavior analytics** that identified revenue spikes during early morning hours (4-5 AM) and on weekends. Additionally, performed **diminishing returns analysis** to optimize spend-response and Return on Ad Spend (ROAS) across various marketing channels

SKILLS AND EXPERTISE

Languages and Frameworks: Python, C++, Javascript, MongoDB, ReactJs, Nodejs, PyTorch, Keras, R*

Additional Skills: NumPy, Pandas, Scikit-Learn, Git, GitHub, Hugging Face, OpenCV*, LangChain*, Unsloth AI* (* elementary proficiency)

COURSEWORK

Programming and Data Structures* | Probability and Statistics | Linear Algebra, Numerical and Complex Analysis | Advanced Calculus | Signals and Systems* | File Organization and Database Systems Lab | Digital Electronics* (* contains Lab component)

EXTRA CURRICULAR ACTIVITIES

- Secured **All India Rank 1460** in Kishore Vaigyanik Protsahan Yojana 2021-22 SA stream under General category, recognized by **Department of Science and Technology, India** for exceptional scientific aptitude and research potential
- Led a team of **8 individuals** as Subhead in **Manga and Anime Society Kharagpur (MASK)** to migrate the official website while maintaining its current version and ensuring task allocation for efficient project completion.
- Managed **Asia's largest** Astrotech fest as Junior Coordinator in **National Students' Space Challenge**, coordinating events, workshops, and competitions to promote space awareness and enhance participant engagement.